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Original Article

Decreases in self-reported sleep duration among U.S. adolescents 2009–2015 and association with new media screen time

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ABSTRACT

Study objectives: Insufficient sleep among adolescents carries significant health risks, making it important to determine social factors that change sleep duration. We sought to determine whether the self-reported sleep duration of U.S. adolescents changed between 2009 and 2015 and examine whether new media screen time (relative to other factors) might be responsible for changes in sleep.

Methods: We drew from yearly, nationally representative surveys of sleep duration and time use among adolescents conducted since 1991 (Monitoring the Future) and 2007 (Youth Risk Behavior Surveillance System of the Centers for Disease Control; total $N = 369{,}595$).

Results: Compared to 2009, adolescents in 2015 were 16%–17% more likely to report sleeping less than 7 h a night on most nights, with an increase in short sleep duration after 2011–2013. New media screen time (electronic device use, social media, and reading news online) increased over this time period and was associated with increased odds of short sleep duration, with a clear exposure–response relationship for electronic devices after 2 or more hours of use per day. Other activities associated with short sleep duration, such as homework time, working for pay, and TV watching, were relatively stable or reduced over this time period, making it unlikely that these activities caused the sudden increase in short sleep duration.

Conclusions: Increased new media screen time may be involved in the recent increases (from 35% to 41% and from 37% to 43%) in short sleep among adolescents. Public health interventions should consider electronic device use as a target of intervention to improve adolescent health.

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Mounting evidence suggests that sufficient sleep duration is crucial for healthy mental and physical functioning. Short sleep duration has been linked to obesity, diabetes, cardiovascular disease, metabolic dysregulation, a variety of psychological and behavioral problems such as depression and substance use, and even increased all-cause mortality [1–4]. Consequences of short sleep are of special importance in pediatric populations for two reasons: sleep patterns established earlier in life shape future sleep habits and health [5,6], and insufficient sleep is especially common in adolescents [7]. Critically, the number of adolescents with insufficient sleep has increased, with short sleep duration becoming increasingly prevalent regardless of age group. Existing evidence supports the existence of a "sleep recession," with only 63% of 15 year olds reporting getting at least 7 h of sleep in 2012, relative to 71.5% in 1991 [8]. However, this and other evidence on changes in sleep over time does not capture the recent revolutionary rise in the use of portable media devices such as smartphones since they saturated the U.S. market around 2012, despite clear association of screen time with curtailed sleep [9].

In response, our goal was to determine whether the trend toward shorter sleep among adolescents has continued in recent years and identify factors possibly responsible for this trend. Multiple studies suggest that screen time is a significant risk factor for short sleep duration [10–12], although there are exceptions [13]. A recent meta-analysis found that adolescents with access to a portable screen device near bedtime had 79% higher odds of sleeping less than 9 h (or 10 h for children) than those without it [14]. This is troubling because by 2011, around 72% of adolescents reported using a cell phone in their bedroom in the

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hour before bedtime [15]. Notably, new media screen time has significantly increased in recent years, especially after smartphones and similar portable devices such as tablets gained greater than 50% market saturation among adolescents around 2011–2012 [16–18]. Portable media devices are of special importance for insufficient sleep as they not only directly displace or delay sleep time by increasing arousal that interferes with sleep but are also easily carried into the bedroom and used in bed before sleep while emitting light that can affect sleep–wake rhythms [19]. These observations raise the key question of whether the last half-decade produced further decreases in sleep among adolescents and whether increased use of new media and/or electronic devices is responsible.

Determining the causes of trends in sleep duration is necessarily difficult as experimental trials are not possible (adolescents cannot be randomly assigned to experience different eras). Ideally, possible causes of temporal changes in sleep duration must meet three criteria. First, they must be correlated with sleep duration across individuals in the population of interest. Second, they must change over time with the trends in sleep duration. Third, other known risk factors for short sleep duration (e.g., homework, working for pay) should not also change at the same time as sleep duration, so these other factors can be discounted as alternative contributing causes. The present analysis examines all three criteria, providing the most stringent test of increased new media screen time as a contributing cause to recent decreases in sleep among adolescents.

1. Research purpose

In the current study, we aimed to examine temporal trends in self-reported short sleep duration (sleeping less than 7 h on most nights) up to 2015 in two large, nationally representative samples of U.S. adolescents conducted since 1991: Monitoring the Future: A Continuing Study of American Youth (MtF) [20,21] and the Youth Risk Behavior Surveillance System (YRBSS) administered by the Centers for Disease Control and Prevention (CDC) [22]. Given the size and representative nature of these databases, they represent the highest likelihood of capturing population trends. We also investigated possible causes for any observed trends by examining how the time adolescents spent on various activities, including new media screen activities (electronic device use, social media use), older media screen activities (TV watching), and non-screen activities (working for pay, exercise, social interaction, homework) affected the odds for short sleep duration and explored whether trends in these activities changed with the trends in sleep duration.

Although a previous study examined trends in sleep duration among adolescents in MtF [8], it ended with data from 2012, just as smartphones gained widespread use among adolescents [16–18]. In addition, this previous study did not include the YRBSS data, another nationally representative sample that has measured sleep duration since 2007 and thus provides an important opportunity for possible replication and confirmation of the MtF survey. To our knowledge, temporal trends in the YRBSS sleep duration data have not been examined (although one study examined the association between shortened sleep and risky behaviors in this dataset [23]). Keyes and colleagues [8] also did not attempt to determine the possible causes for the decline in sleep duration; they in fact specifically suggested that future research investigate such possible mechanisms, noting that "further research unraveling such effects is critical to inform public health approaches to curbing the trends of decreased sleep among adolescents" (p. 6). To that end, we investigate relationships between both screen activities and non-screen activities on short sleep duration.

2. Method

2.1. Sample

The Monitoring the Future (MtF) survey, funded by the National Institute on Drug Abuse, collects nationally representative samples of 8th, 10th, and 12th graders (see http://www.monitoringthefuture.org). All procedures of the survey are annually reviewed and approved by the University of Michigan Institutional Review Board. The participation rate of schools is between 66% and 80%, and the student participation rate is between 79% and 83% [24]. About 15,000 students in each grade are sampled each year in the spring. We examined the data from 1991, when the survey was administered to students in all three grades, until 2015, the most recent data available. For the current analyses, N (1991–2015) = 301,980; N (2009–2015) = 75,784.

The YRBSS, administered by the CDC, has surveyed 9th, 10th, 11th, and 12th graders every other year since 1991, with data available up to 2015 [25]. The YRBSS protocol was approved by the CDC Institutional Review Board. The YRBSS surveys a nationally representative sample of U.S. high school students (including both public and private schools) using a three-stage cluster sample design. Average school, student, and overall response rates are 78%, 86%, and 71%, respectively. The data can be weighted to make the results nationally representative and control for student nonresponse [25]. The sleep question was asked beginning in 2007: N (2007–2015) = 67,615; N (2009–2015) = 55,576.

Both these datasets use a time-lag design, surveying like-age samples in different years. This allows the separation of age effects from the effects of time period and cohort. However, participants are assessed only once and not followed as in a longitudinal design, precluding analyses of within-subject effects (such as the time sequence of activities and sleep duration).

2.2. Measures

MtF asks, "How often do you get at least 7 hours of sleep?" with response choices of "never," "seldom," "sometimes," "most days," "nearly every day," and "every day." [24] Those who answered never, seldom, or sometimes were counted as not getting 7 or more hours of sleep on most days.

Since 2007, YRBSS has asked, "On an average school night, how many hours of sleep do you get?" with response choices of "4 or less hours," "5 hours," "6 hours," "7 hours," "8 hours," "9 hours," and "10 or more hours." [26] Those who answered 4 or less hours, 5 h, or 6 h were counted as not getting 7 or more hours of sleep on most days.

Each survey also included items on new media screen activities (electronic device use, social media use, reading news online), older media screen activities (TV watching), and non-screen activities (exercise, in-person social interaction, homework, paid jobs) asked of the same participants. Some of these items measure time spent in hours and others in general increments (see Supplemental Material). Social media use was only asked since 2009; therefore, for the analyses examining the associations of screen vs. non-screen activities with sleep, we focused on the period 2009–2015 when all the activities items were asked.

2.3. Data analysis

We examined odds ratios for each of the activities on short sleep duration, examining whether individuals who engaged in the activity were more or less likely to sleep less than 7 h on most nights. We also report these odds ratios after controlling for grade, race, and sex. Because relative risk is easier to interpret, we also report some findings in the text as relative risk. As the survey designs

include different participants every year (a time-lag design), we were only able to examine concurrent odds ratios between activities and sleep; thus, lagged analyses examining which increased first (such as those used in longitudinal designs following the same participants) were not possible. We also examined the exposure–response relationship between new media screen time and short sleep duration, relying on the YRBSS data as new media screen time was measured in hours of the same participants also asked about sleep duration only in that survey. We *Z*-scored the activity variables within 2009–2015 to present them in the same figure.

3. Results

The number of adolescents getting insufficient sleep abruptly increased after 2011–2013. By 2015, more than 40% of adolescents did not get 7 or more hours of sleep on most nights across both datasets (see Fig. 1). Between 2009 and 2015, 16%–17% more adolescents reported insufficient sleep (with increases from 35% to 41% in MtF and from 37% to 43% in YRBSS). This is in addition to the previously reported increase in short sleep duration since 1991, when only 26% of adolescents got insufficient sleep; thus, 58% more adolescents reported insufficient sleep in 2015 compared to 1991.

We then examined possible reasons for the increase. New media screen activities were associated with increased odds for short sleep duration. Not all screen media had a pronounced negative association; odds ratios for TV watching were weaker and less consistent. Spending more than 10 h a week on homework and working at a paid job also increased the odds for short sleep duration. However, activities such as exercise and in-person social interaction instead were associated with decreased odds for short sleep duration (see Table 1).

Only some of these activities displayed clear trends over time and thus might plausibly be responsible for the sudden rise in short sleep duration between 2009 and 2015 (see Fig. 2a–b and Table 2). The only activities that were both (A) associated with increased odds of short sleep duration and (B) changed more than one-tenth of a standard deviation unit ($d = 0.10$) between 2009 and 2015 were the three new media screen activities: electronic device use ($d = 0.39$, increasing 44 min a day), social media use ($d = 0.43$), and Internet news consumption ($d = 0.33$). Homework time was associated with increased odds of short sleep duration but increased only 4 min a day, compared to the increase of 44 min a day in electronic device use; in addition, homework time was unchanged 2012–2015 when sleep duration increased the most. Working for pay and TV watching both decreased over this time period. Thus, at least among the activities measured here, increased new media screen time is the most plausible cause of shortened sleep duration among adolescents since 2010.

Electronic device use appears to have a fairly linear exposure–response relationship with short sleep duration (see Fig. 3). Hours of electronic device use were positively correlated with failing to get 7 or more hours of sleep on most nights,

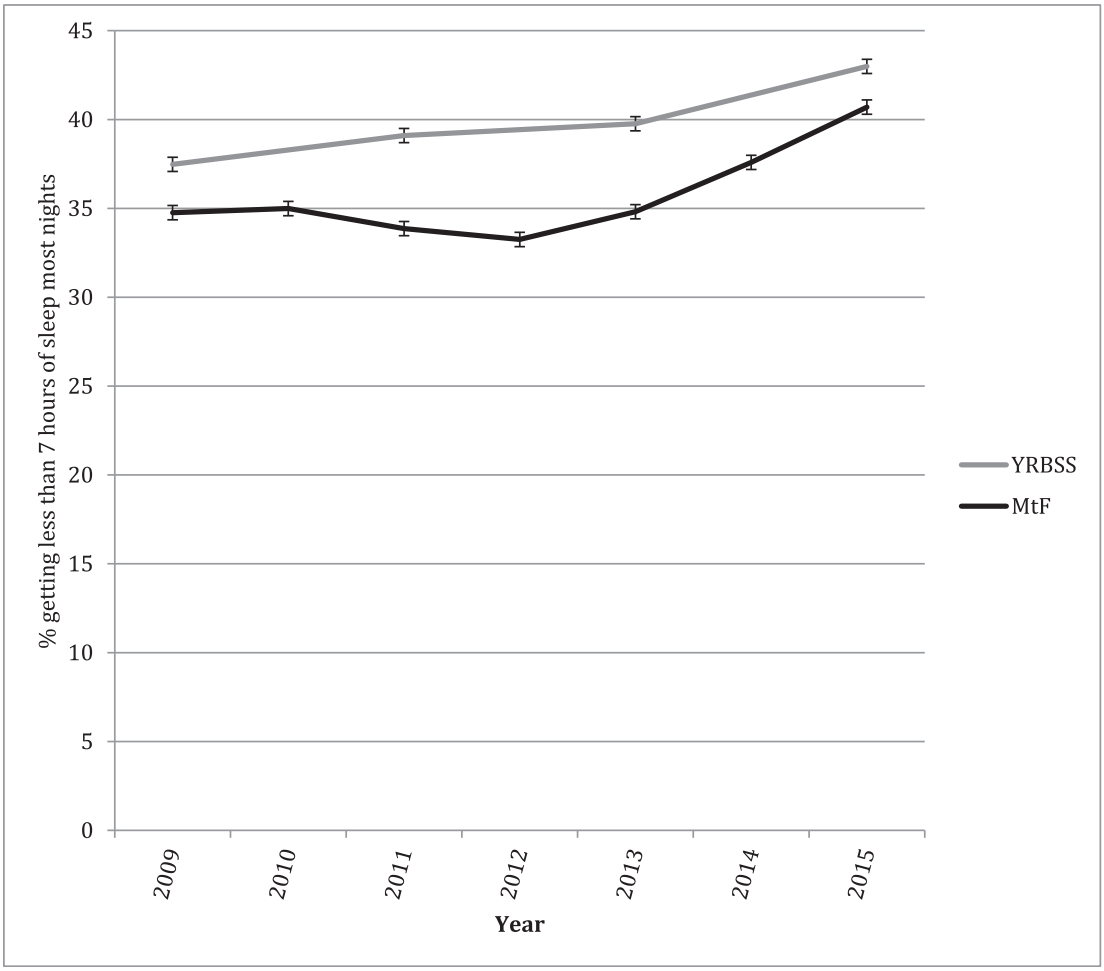


Fig. 1. Percentage of U.S. adolescents reporting getting less than 7 h of sleep on most nights: 8th, 10th, and 12th graders, Monitoring the Future (MtF) and 9th–12th graders, Youth Risk Surveillance System (YRBSS), 2009–2015. Note: Error bars represent ±1 SE.

Table 1
Odds ratios for getting less than 7 h of sleep for screen and non-screen activities (2009–2015) and for birth cohort/time period in U.S. adolescents.

	Bivariate odds ratio for getting less than 7 h of sleep	With controls for sex, race, SES, and grade
Screen activities		
Electronic device use 3+ h a day (YRBSS)	1.52 (1.47, 1.58)	1.56 (1.50, 1.62)
Social media every day (MtF)	1.31 (1.27, 1.36)	1.16 (1.11, 1.21)
Internet news every day (MtF)	1.18 (1.14, 1.22)	1.15 (1.10, 1.21)
TV viewing 3+ h a day (MtF)	1.05 (1.01, 1.09)	1.06 (1.01, 1.11)
TV viewing 3+ h a day (YRBSS)	1.05 (1.01, 1.09)	0.99 (0.96, 1.03)
Non-screen activities		
In-person social interaction above average (MtF)	0.92 (0.89, 0.95)	0.91 (0.88, 0.94)
Exercise most days (MtF)	0.51 (0.49, 0.52)	0.53 (0.51, 0.55)
Sports or exercise 60+ minutes a day 5+ days a week (YRBSS)	0.66 (0.63, 0.69)	0.72 (0.69, 0.76)
Homework 10+ hours a week (MtF)	1.17 (1.13, 1.21)	1.13 (1.08, 1.18)
Having a paid job (MtF)	1.22 (1.18, 1.26)	1.07 (1.03, 1.12)
Birth cohort/time period		
1991 vs. 2015 MtF	1.96 (1.86, 2.07)	1.93 (1.80, 2.06)
2007 vs. 2015 YRBSS	1.19 (1.14, 1.25)	1.18 (1.12, 1.24)
2009 vs. 2015 MtF	1.29 (1.22, 1.36)	1.33 (1.25, 1.42)
2009 vs. 2015 YRBSS	1.26 (1.20, 1.32)	1.24 (1.19, 1.30)

Notes: 1. 95% CI range for the odds ratio shown in parentheses.
2. Internet news and TV viewing were only asked of 8th and 10th graders in MtF.
3. YRBSS analyses are not controlled for SES as this is not measured in the dataset.

$r(67{,}175) = 0.11$, $p < 0.001$. (This correlation was relatively constant across the years; it was 0.10 in every year but 2013, when it was 0.12). In a regression equation including electronic device use (centered) and its quadratic term, the linear relationship (Beta = 0.08, $p < 0.001$) was similar to the curvilinear relationship (Beta = 0.07, $p < 0.001$). As Fig. 3 shows, the associations between electronic device use and sleep appear after two or more hours of use per day and grow steadily larger. Thirty-five percent of teens using electronic devices 1 h a day slept less than 7 h a night, whereas half (52%) of teens using electronic devices 5 or more hours a day slept this little. Thus, those who used electronic devices 5 or more hours a day were 50% more likely to experience short sleep duration than those spending 1 h a day on these devices.

4. Discussion

More U.S. adolescents in recent years failed to get 7 or more hours of sleep on most nights, with an abrupt increase in short sleep duration after 2011–2013. By 2015, more than 40% of adolescents reported getting insufficient sleep. Most research suggests that adolescents ages 12 to −17 require an average of 9 h a night of sleep. Thus, more adolescents are sleep deprived.

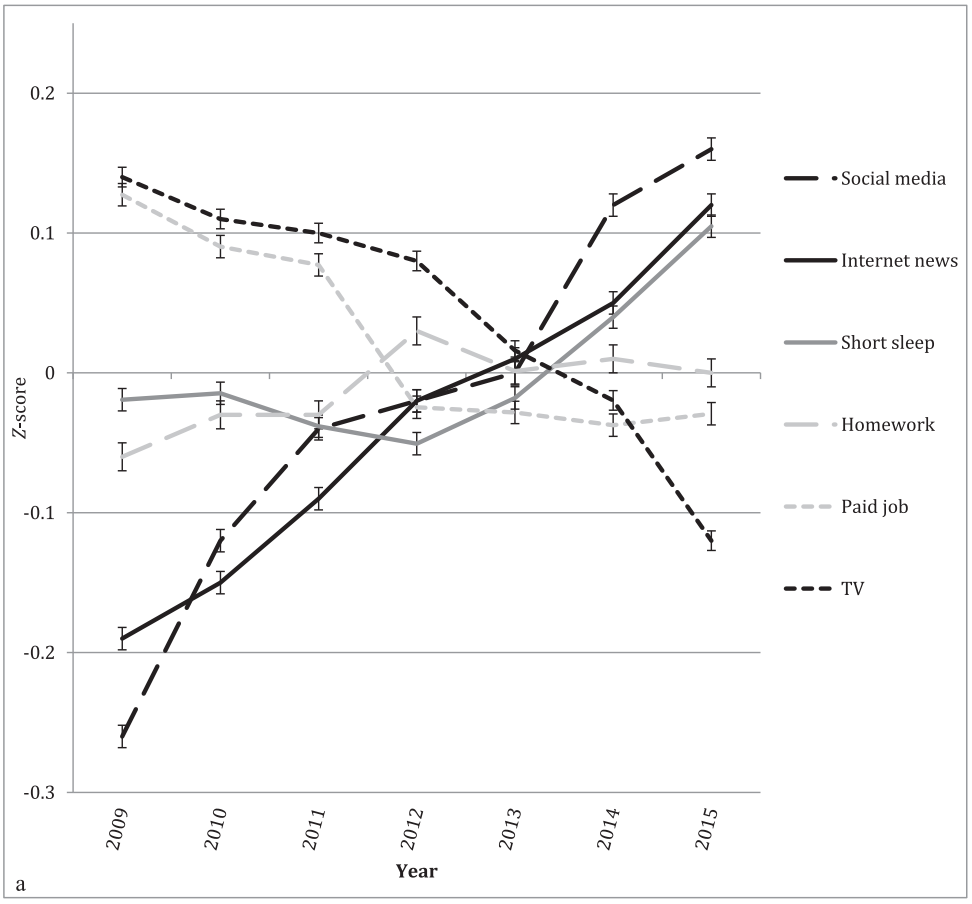
New media screen time (including reported use of electronic devices, social media, and reading news online) significantly increased the likelihood of short sleep duration, and time spent on these screen activities increased between 2009 and 2015. This may be due to the increasing prevalence of smartphones during that time period. TV watching had weaker or trivial associations with sleep duration and also decreased over this time, making it unlikely that it caused the recent increase in short sleep duration. Other factors that increased the odds of shorter sleep duration across cohorts, such as homework and working for pay, either changed little or decreased during these years, also lowering the probability that they influenced the changes in sleep. Thus, among the activities measured here, increase in new media screen time is the most likely culprit for the abrupt increase in short sleep duration after 2011–2013. Electronic device use of 2 or more hours a day has a fairly linear exposure–response association with sleep; the average adolescent spends 2 h and 25 min a day on electronic devices. Barring experimentation, this is the strongest evidence to date that the use of electronic devices among adolescents is responsible for more adolescents obtaining insufficient sleep.

Short sleep duration also increased sharply during the 1990s [8]; however, it is more difficult to determine the causes of that trend because of limitations in measurement. MtF did not measure computer gaming until 2008 (and even then asked it of different participants than the sleep item). Electronic device use (which includes computer gaming) was not measured in the YRBSS database until 2003. Thus it is difficult to determine whether increased use of computers and other electronic devices (such as game consoles) during the 1990s were responsible for the increase in short sleep duration over that decade, a likely possibility given the rise of the Internet during that time. Homework time, time spent working for pay, and TV watching all declined during the 1990s in the MtF data [18] and thus are unlikely to be the cause.

These data are limited in that sleep duration is retrospectively self-reported. First, the format of the question did not allow for precise estimation of actual sleep duration, although it did enable us to generate comparable categories of shortened sleep between the two different surveys. This reduces the precision of the data and suggests that our analyses were relatively conservative. Second, previoius research found only a moderate convergence between self-reported and actigraphically recorded sleep duration, with especially low convergence among those distressed about their sleep or health [27]. Future research should aim to address screen time while assessing sleep through methods other than self-report. Finally, equivalent chronological age across cohorts may not reflect equivalent "biological" age because of changing biological and social factors advancing the onset of puberty [28]. However, the sudden increase in short sleep duration after a decade of relative stability during the 2000s contrasts with the linear decrease in age at onset of puberty, suggesting there are likely other factors at work. In addition, the measure of electronic device use is limited as the measure changed its wording to include smartphones and tablets in 2013. Although this more inclusive wording better captures the electronic device use of today's adolescents, the increase in electronic device use 2009–2015 shown in Fig. 2b may be smaller. However, the correlation between electronic device use and short sleep duration was fairly constant in each year. Furthermore, the wordings of the other activity items and sleep duration items were constant over time.

4.1. Research implications

Although these results support a possible causal inference that increased new media screen time is responsible for decreases in sleep duration over time, they do not isolate specific processes responsible for this influence. Multiple processes have been identified as potential mechanisms responsible for the erosive impact of screen time on sleep, including displacement of sleep time, increases in arousal that harm sleep quality, re-entrainment of circadian rhythms due to light exposure, and increases in depressive symptoms [7,15,19,29]. Given the presently documented cumulative impact of extensive screen time fueled by smartphone technology, it is imperative to evaluate the relative importance of these processes in undermining adolescent sleep and to identify how specific pathways of influence combine. These pathways are



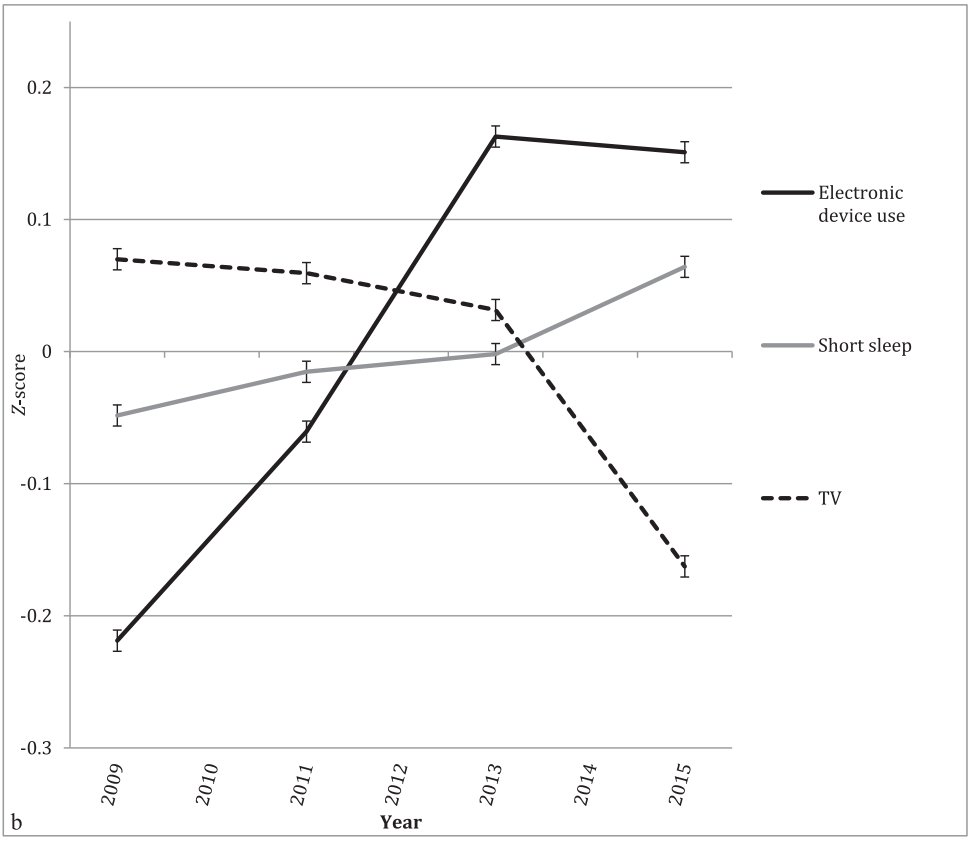


Fig. 2. a: Temporal trends in screen activities (black), selected non-screen activities (light gray), and self-reported short sleep duration (dark gray), 8th, 10th, and 12th graders, Monitoring the Future, 2009–2015. NOTE: Error bars represent ±1 SE. b: Temporal trends in time spent on electronic devices, time spent watching TV, and self-reported short sleep duration, Youth Risk Behavior Surveillance System, 9th–12th graders, 2009–2015. Note: Error bars represent ±1 SE.

Table 2
U.S. adolescents' activities and short sleep duration, 2009–2015.

	N	2009	2010	2011	2012	2013	2014	2015	*d* 2009 to 2015
Electronic device use hours per day (YRBSS)	59,730	1.68 (1.72)	–	2.00 (1.85)	–	2.44 (2.13)	–	2.42 (2.14)	0.39
Social media use, 1–5 scale (MtF)	137,029	3.98 (1.42)	4.15 (1.34)	4.25 (1.30)	4.27 (1.25)	4.30 (1.24)	4.45 (1.17)	4.51 (1.12)	0.43
Internet news use, 1–5 scale (MtF)	149,632	4.34 (1.02)	4.38 (1.01)	4.43 (0.97)	4.50 (0.90)	4.53 (0.87)	4.56 (0.87)	4.62 (0.80)	0.33
TV viewing hours per day (MtF)	149,696	2.41 (1.61)	2.36 (1.61)	2.35 (1.59)	2.20 (1.55)	2.21 (1.59)	2.15 (1.60)	1.99 (1.56)	−0.27
TV viewing hours per day (YRBSS)	59,623	2.06 (1.72)	–	2.04 (1.74)	–	2.00 (1.77)	–	1.66 (1.65)	−0.24
In-person social interaction *Z*-score average of 6 activities (MtF)	139,477	0.14 (0.62)	0.12 (0.64)	0.08 (0.62)	0.03 (0.61)	−0.02 (0.61)	−0.06 (0.61)	−0.08 (0.60)	−0.36
Exercise/sports, 1–5 scale (MtF)	165,750	3.98 (1.31)	3.97 (1.32)	4.04 (1.28)	4.04 (1.29)	3.98 (1.33)	3.96 (1.36)	3.95 (1.36)	−0.02
Homework hours per day (MtF)	160,742	0.85 (0.90)	0.88 (0.94)	0.88 (0.93)	0.94 (0.98)	0.91 (0.97)	0.92 (0.98)	0.92 (0.97)	0.07
Having a paid job % yes (MtF)	162,190	0.32 (0.47)	0.30 (0.46)	0.30 (0.46)	0.25 (0.43)	0.25 (0.43)	0.25 (0.43)	0.25 (0.43)	−0.16
Short sleep % yes (MtF)	75,784	0.35 (0.48)	0.35 (0.48)	0.34 (0.47)	0.33 (0.47)	0.35 (0.48)	0.38 (0.48)	0.41 (0.49)	0.12
Short sleep % yes (YRBSS)	55,576	0.37 (0.48)	–	0.39 (0.49)	–	0.40 (0.49)	–	0.43 (0.50)	0.12

Note: Differences are shown in *d* (difference in standard deviations).

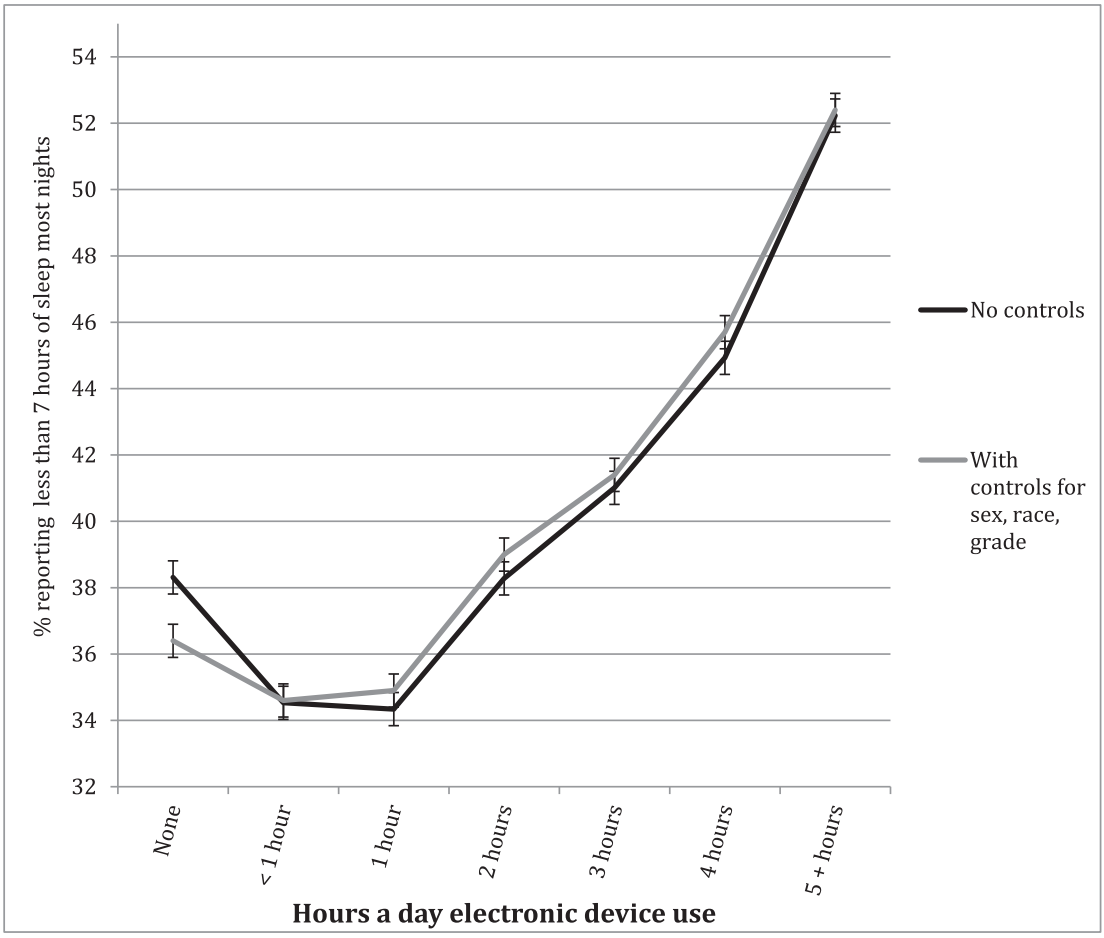


Fig. 3. Exposure–response relationship between electronic device use and self-reported short sleep duration, Youth Risk Behavior Surveillance System, 2007–2015. Notes: 1. Error bars represent ±1 SE, 2. $n = 67{,}175$.

likely to involve conscious choices driven by personal goals (e.g., delaying bedtime to chat with a friend), stress and arousal (e.g., disrupted sleep due to online arguments or feelings of depression), and circadian biological responses (e.g., shifts in circadian phase due to light exposure).

4.2. Policy implications

These findings also carry critical implications for public health and policy. Much of the information consumed in modern society (including some formal education) is delivered through screens, especially through personal tablet or smartphone devices. To avoid interfering with students' sleep, policymakers need to consider whether educational curricula and student evaluation methods require screen use and how much flexibility to afford students for *when* to use mobile screens. It is also important to track and monitor screen exposure in pediatric populations to identify those most at risk for sleep disruption and prevent potential development of poor sleep hygiene and chronic sleep problems. The data here suggest that adolescents using electronic devices 2 h a day or

more, and especially those using them 4 h a day or more, are most in need of intervention. Because screen time is an important risk factor, institutions need to consider policies that regulate both the quantity and quality of screen time among children and adolescents.

5. Conclusion

The last decade witnessed unprecedented shifts in the adoption and use of new electronic media devices, especially among children and adolescents. Scientific evidence on the impact of this technology is only now accumulating, and the findings presented here implicate the use of electronic devices and new media in abrupt and significant decreases in sleep time. Both individuals and institutions should consider ways to mitigate the unwanted effects of electronic devices on sleep and health.

Conflict of interest

The ICMJE Uniform Disclosure Form for Potential Conflicts of Interest associated with this article can be viewed by clicking on the following link: https://doi.org/10.1016/j.sleep.2017.08.013.

Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.sleep.2017.08.013.

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